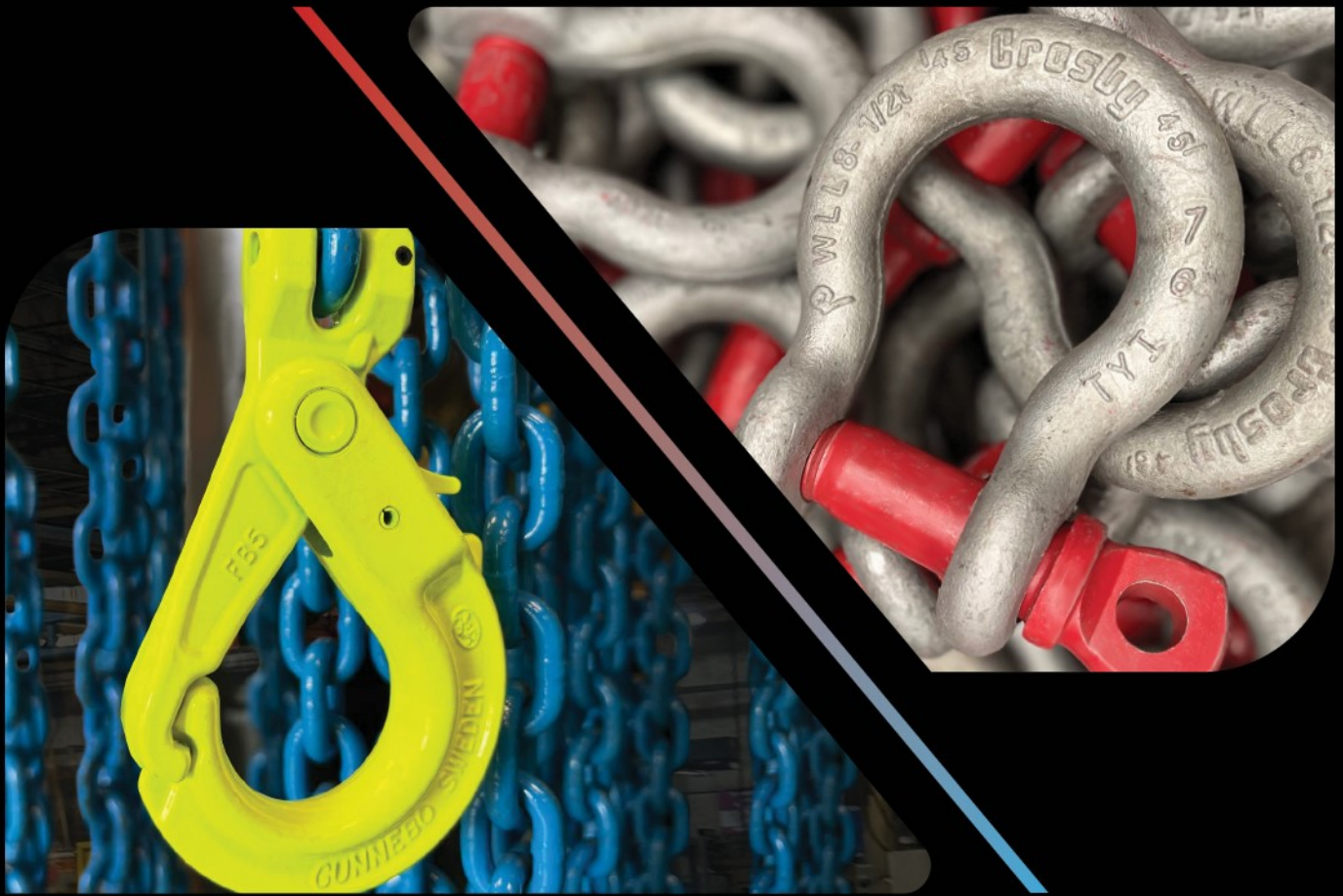




MASTER LINKS





Crosby®



GUNNEBO®
Industries

PRODUCT CATALOG

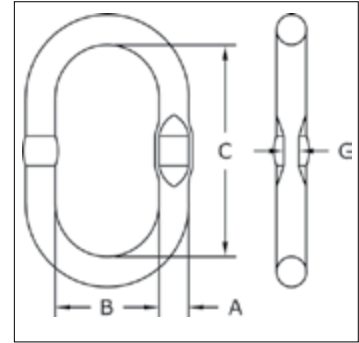
with application & warning information

Imperial

A-344



- Alloy steel — Quenched & Tempered.
- Individually Proof Tested to values shown, with certification.
- Design Factor of 5 to 1.
- Proof Tested with 70% inside width special fixtures sized to prevent localized point loading per EN1677.
- Each main link is marked with Product Identification Code (PIC) for material traceability, Grade, CE, chain size and the "CG" (Crosby Group).
- A-344 master links are type approved to DNV Certification Notes 2.7-1- Offshore Containers. These Crosby master links are 100% proof tested. Every batch is impact tested. The tests are conducted by Crosby and 3.1 test certification is available upon request.
- Engineered Flat for use with S-1325A coupler link.
- Fatigue rated to 20,000 cycles at 1.5 times the Working Load Limit.
- Meets or exceeds all requirements of ASME B30.26 including identification, ductility, design factor, proof load and temperature requirements. Importantly, these links meet other critical performance requirements including fatigue life, impact properties and material traceability, not addressed by ASME B30.26.
- Available only in EMEA.



7/16" through 1-7/32" have Engineered Flat.

Grade 80 A-344 Welded Master Links available with Engineered Flat

Stock No.	Weight Each (lb)	Grade 100 Chain Sling		Grade 80 Chain Sling		WLL (lb)	Proof Load (lb)	Dimensions (in)				Engineered Flat Size for S-1325A (in)
		Single Leg Chain Size (in)	Double Leg Chain Size (in)	Single Leg Chain Size (in)	Double Leg Chain Size (in)			A	B	C	G	
1256988	0.8	6mm, 9/32	6mm	6mm, 9/32	6mm, 9/32, 5/16	7,000	17,632	0.51	2.36	4.72	0.26	6mm, 9/32, 5/16
1257002	1.9	5/16, 3/8	9/32	5/16, 3/8	5/16	9,000	22,701	0.67	3.54	6.30	0.33	3/8
1257072	2.3	3/8, 1/2	5/16	3/8, 1/2	3/8	14,700	37,027	0.75	3.54	6.30	0.33	3/8, 1/2
1257268	5.2	3/8, 1/2	3/8	3/8, 1/2	3/8	15,400	38,570	0.87	5.71	10.83	0.41	1/2
1257212	3.6	3/8, 1/2	3/8	5/8	3/8	19,400	48,488	0.87	3.94	7.09	0.41	1/2
1257332	6.7	1/2	-	1/2, 5/8	3/8	19,600	48,929	0.98	5.71	10.83	0.53	5/8
1257282	5.3	5/8, 1/2	3/8	5/8	1/2	25,300	63,475	0.98	4.53	8.27	0.53	5/8
1257382	8.5	5/8, 1/2	1/2	5/8	1/2	28,600	71,630	1.10	5.71	10.83	0.53	5/8
1257422	10.6	5/8, 3/4	1/2	3/4	5/8	37,400	93,670	1.26	5.71	10.83	0.66	-
1257492	15.2	3/4	5/8	3/4, 7/8	3/4	52,900	132,240	1.42	6.10	11.22	-	-
1257502	16.1	7/8	3/4	7/8	7/8	69,400	173,675	1.57	5.51	10.63	-	-
1257562	28.4	1	7/8	1	1	84,400	210,923	1.77	7.09	13.39	-	-
1257632	42.1	1, 1-1/4	7/8	1	1	99,200	247,950	2.01	8.46	15.35	-	-
1257573	55.3	1-1/4	1	1-1/4	1-1/4	147,600	369,170	2.17	7.99	15.98	-	-

5:1 Design Factor. Applications with wire rope and synthetic sling generally require a Design Factor of 5. Based on single leg sling (in-line load), or resultant load on multiple legs with an included angle less than or equal to 120 degrees. Proof Test Load equals or exceeds the requirement of ASTM A952(8.1) and ASME B30.9. Chain slings require that the Design Factor be 4:1. Refer to applications & warnings to determine product's actual Ultimate Load. There are no manufactured flats on links over 1 1/4" (32mm). Two largest sizes are available globally.

Fatigue Rated

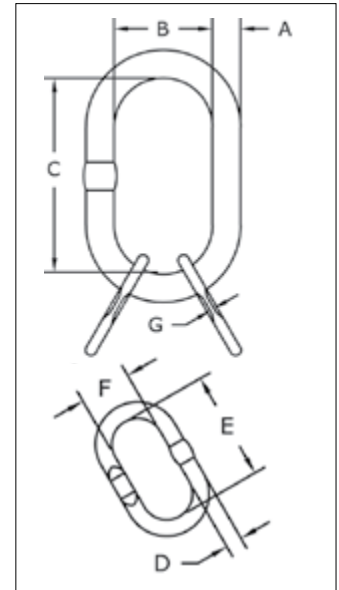


APPLICATION AND WARNING INFORMATION
SECTION 17

A-347



- Alloy steel — Quenched & Tempered.
- Individually Proof Tested to values shown, with certification.
- Design Factor of 5 to 1.
- Proof Tested with 70% inside width special fixtures sized to prevent localized point loading per EN1677.
- Each main link is marked with Product Identification Code (PIC) for material traceability, Grade, CE, chain size and "CG." Each sublink is marked with traceability code.
- A-347 master links are type approved to DNV Certification Notes 2.7-1- Offshore Containers. These Crosby master links are 100% proof tested. Every batch is impact tested. The tests are conducted by Crosby and 3.1 test certification is available upon request.
- Engineered Flat for use with S-1325A coupler link.
- Fatigue rated to 20,000 cycles at 1.5 times the Working Load Limit.
- Meets or exceeds all requirements of ASME B30.26 including identification, ductility, design factor, proof load and temperature requirements. Importantly, these links meet other critical performance requirements including fatigue life, impact properties and material traceability, not addressed by ASME B30.26.
- Available only in EMEA (Europe, Middle East, and Africa).



5

Grade 80 A-347 Welded Master Link Assembly with Engineered Flat

Stock No.	Weight Each (lb)	Grade 100 Chain Sling Three / Four Legs Chain Size (in)	Grade 80 Chain Sling Three / Four Legs Chain Size (in)	WLL (lb)	Proof Load (lb)	Dimensions (in)							Engineered Flat Size for S1325A Chain Size (in)
						A	B	C	D	E	F	G	
1257755	2.4	-	6mm	7,000	17,632	0.51	2.36	4.72	0.51	4.72	2.36	0.26	6mm
1257762	3.5	6mm	6mm, 9/32	9,000	22,701	0.67	3.54	6.30	0.51	4.72	2.36	0.26	6mm
1257832	3.9	6mm	9/32	9,200	23,362	0.75	3.54	6.30	0.51	4.72	2.36	0.26	9/32
1258058	7.3	5/16, 9/32	5/16	15,400	38,570	0.87	3.94	7.09	0.67	6.30	3.54	0.33	3/8
1258067	8.9	5/16, 9/32	5/16	15,400	38,570	0.87	5.71	10.83	0.67	6.30	3.54	0.33	3/8
1258049	8.4	5/16	3/8	18,700	46,725	0.87	3.94	7.09	0.75	6.30	3.54	0.33	3/8
1258076	10.1	5/16	3/8	19,600	49,149	0.98	4.53	8.27	0.75	6.30	3.54	0.33	3/8
1258102	11.4	5/16	3/8	19,600	49,149	0.98	5.71	10.83	0.75	6.30	3.54	0.33	3/8
1258142	15.6	3/8	1/2	31,900	80,005	1.10	5.71	10.83	0.87	7.09	3.94	0.41	1/2
1258182	21.2	3/8	1/2	37,400	93,670	1.26	5.71	10.83	0.98	8.27	4.53	0.53	5/8
1258185	28	1/2	5/8	52,000	130,036	1.42	6.10	11.22	1.10	7.48	4.33	0.53	5/8
1258187	40.6	5/8	5/8	61,900	154,941	1.57	5.51	10.63	1.26	10.83	5.71	0.66	-
1258402	58.6	5/8	3/4	84,400	211,143	1.77	7.09	13.39	1.42	11.22	6.10	-	-
1258471	78.2	3/4	7/8	99,200	247,950	2.01	8.46	15.35	1.57	10.63	5.51	-	-
1258491	134.6	7/8	1	147,600	369,170	2.17	7.99	15.98	2.01	15.35	8.46	-	-

5:1 Design Factor. Applications with wire rope and synthetic sling generally require a Design Factor of 5. Based on single leg sling (in-line load), or resultant load on multiple legs with an included angle less than or equal to 120 degrees. Proof Test Load equals or exceeds the requirement of ASTM A952(8.1) and ASME B30.9. Chain slings require that the Design Factor be 4:1. Refer to applications & warnings to determine product's actual Ultimate Load. There are no manufactured flats on links over 1 1/4" (32mm).

Fatigue Rated

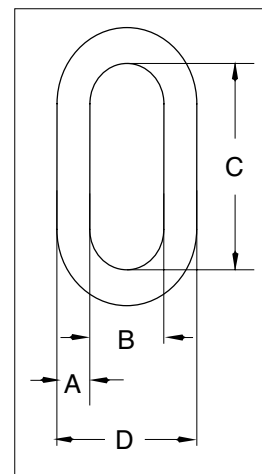


APPLICATION AND WARNING INFORMATION
SECTION 17

G-340 / S-340



- Forged carbon steel - Quenched & Tempered
- Self colored (S) or hot-dip galvanized (G).



G-340/S-340 Weldless End Links

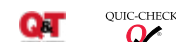
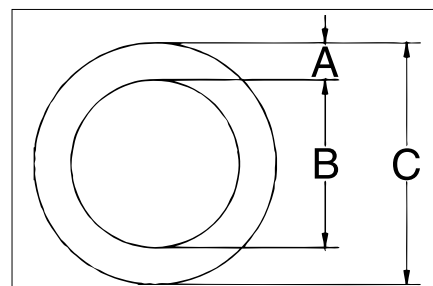
Size (A) (in)	Stock No.		Working Load Limit (lb)	Weight Each (lb)	Dimensions (in)			
	G-340 Galv.	S-340 S.C.			A	B	C	D
5/16	1014057	1014066	2,500	.15	.31	.50	1.75	1.18
3/8	1014075	1014084	3,800	.22	.38	.56	1.88	1.38
1/2	1014093	1014100	6,500	.49	.50	.75	2.38	1.81
5/8	1014119	1014128	9,300	.97	.63	1.00	3.25	2.32
3/4	1014137	1014146	14,000	1.51	.75	1.13	3.50	2.68
7/8	1014155	1014164	12,000	2.59	.88	2.00	5.13	3.75
1	1014173	1014182	15,200	3.95	1	2.25	5.75	4.25
1-1/4	1014191	1014208	26,400	7.30	1.25	2.50	7.00	5.00
1-3/8	1014217	1014226	30,000	10.38	1.38	2.75	7.75	5.50

5:1 Design Factor. Based on single leg sling (in-line load), or resultant load on multiple legs with an included angle less than or equal to 120°.

S-643



- Forged carbon steel - Quenched & Tempered.



S-643 Weldless Rings

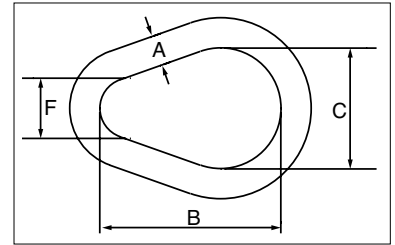
Size (in)	Stock No	Working Load Limit Single Pull (lb)	Weight Each (lb)	Dimensions (in)		
				A	B	C
7/8 x 4	1013780	7,200	2.72	.88	4.00	5.75
7/8 x 5-1/2	1013806	5,600	3.47	.88	5.50	7.25
1 x 4	1013824	10,800	3.69	1.00	4.00	6.00
1-1/8 x 6	1013842	10,400	6.60	1.13	6.00	8.25
1-1/4 x 5	1013860	17,000	6.82	1.25	5.00	7.50
1-3/8 x 6	1013888	19,000	10.12	1.38	6.00	8.75

6:1 Design Factor.

A-341



- Forged alloy steel — Quenched & Tempered.
- Individually Proof Tested at 2 times Working Load Limit with certification.



A-341 Alloy Pear Shaped Links

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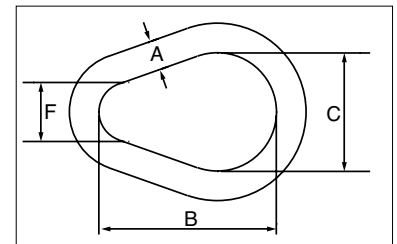
Size (A) (in)	Stock No	Working Load Limit		Weight Each (lb)	Dimensions (in)		
		(lb)	(t)		B	C	F
1/2	1013575	7,000	3.15	.55	3.00	2.00	1.00
5/8	1013584	9,000	4.09	1.10	3.75	2.50	1.25
3/4	1013595	12,300	5.59	1.76	4.50	3.00	1.50
7/8	1013604	15,000	6.81	2.82	5.25	3.50	1.75
1	1013613	24,360	11.0	4.22	6.00	4.00	2.00
1 1/8	1013622	30,600	13.9	6.25	6.50	4.50	2.25
1 1/4	1013631	36,000	16.4	8.25	7.75	5.00	2.50
1 3/8	1013640	43,000	19.5	11.25	8.25	5.50	2.75
1 1/2	1013654	54,300	24.7	14.25	9.00	6.00	3.00
1 3/4	1013672	84,900	38.6	22.50	10.50	7.00	3.50
2	1013690	102,600	46.6	34.00	12.00	8.00	4.00

5:1 Design Factor. Based on single leg sling (in-line load), or resultant load on multiple legs with an included angle less than or equal to 120°.

G-341 / S-341



- Forged carbon steel - Quenched & Tempered.
- Self colored (S) or hot-dip galvanized (G).



G-341 / S-341 Weldless Sling Links

Size (A) (in)	Stock No.		Working Load Limit Single Pull (lb)	Weight Each (lb)	Dimensions (in)		
	G-341 Galv.	S-341 S.C.			B	C	F
3/8	1013897	1013904	1,800	.23	2.25	1.50	.75
1/2	1013913	1013922	2,900	.55	3.00	2.00	1.00
5/8	1013931	1013940	4,200	1.06	3.75	2.50	1.25
3/4	1013959	1013968	6,000	1.88	4.50	3.00	1.50
7/8	1013977	1013986	8,300	2.75	5.25	3.50	1.75
1	1013995	1014002	10,800	4.35	6.00	4.00	2.00
1 1/4	1014011	1014020	16,750	7.60	7.75	5.00	2.50
1 3/8	1014039	1014048	20,500	11.30	8.25	5.50	2.75

6:1 Design Factor. Based on single leg sling (in-line load), or resultant load on multiple legs with an included angle less than or equal to 120°.

Identification of our Master Links

To provide good readability and traceability our master links have the following marking:

Product type

- M - represents single type master link.
- MT - represents master link assembly.
- MF - represents single type master link with engineered flat.
- MFH - represents master link with engineered flat and DIN style crane hooks.
- MFX - represents enlarged single type master link with engineered flat.

Size designation

- The size is linked to the WLL as well as to compatible products, like attachment couplers and other components.
- Trade size.
- The size expressed in inch.

Approved by BG/DGUV

- H32 – represents Gunnepro Industries' manufacturing ID. The ID also represents a 3rd part audit by BG in Germany.

Traceability code

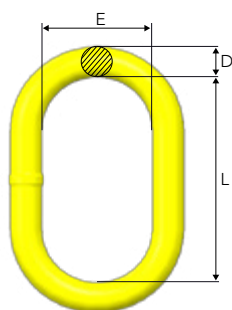
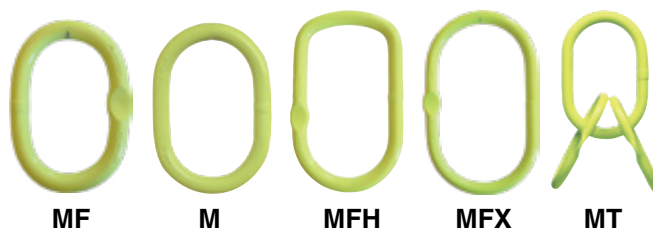
- The traceability code is unique for the production batch and consists of 3 letters representing the year of manufacture, the factory, and the production run, for example BVB. The traceability code makes it possible to trace and track the product through the whole production process back to the raw material used for the actual product.

Gunnepro Sweden

- To clearly highlight the Gunnepro Industries brand, our master links are marked with Gunnepro, Sweden.

Meets the standards

- The markings fulfills the requirements of EN 1677-4, ASTM A952 and AS 3775.2.

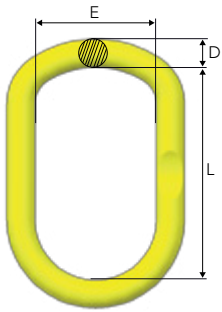


- Made from Quenched & Tempered, Fine Grain Alloy Steel from European Steel Mills.
- Links are individually Proof Tested to values shown, with certification.
- Proof Test with max 60% inside width fixture to prevent localized point loading per ASTM A952.
- Each link is marked with Product Identification Code (PIC) for traceability, Grade, chain size, SWE, GUNNEBO and BG/DGUV manufacturing ID (H32).
- Fatigue rated to at least 20,000 cycles at 1.5 times the Working Load Limit.
- Designed for use with chain, wire or synthetic rope. Applications with wire and synthetic rope generally require a 5:1 Design Factor.
- 3/4-leg requires Gunnepro Industries CG, CGD, CL or CLD components. Engineered Flat compatible with Crosby S-1325 and Gunnepro Industries BL Omega Link.
- Fulfills or exceeds requirements in EN1677:2008, ASTM A952/A952M-02, AS 3775:2014 and AS 3776:2015.

Grade 100 Welded Master Link M and Welded Master Link MF with Engineered Flat

Stock No.	Code	WLL (lb)	Proof Load (lb)	For Grade 100 Chain Size (in)			Dimensions (in)				Weight (lb)
				1-leg	2-leg	3/4-leg	L	E	D	Flat Thickness	
Z200060	MF-6-10	4,408	11,016	7/32"	-	-	3.94	2.36	0.43	7/32"	0.44
Z200086	MF-86-10	7,053	17,536	5/16"	7/32"	-	4.70	2.36	0.51	7/32"	0.88
Z200108	MF-108-10	11,902	29,676	3/8"	5/16"	7/32"	6.30	3.54	0.67	5/16"	1.9
Z2111310	MF-1310-10	18,073	45,189	1/2"	3/8"	5/16"	7.09	3.94	0.87	3/8"	3.5
Z211613	MF-1613-10	29,974	75,090	5/8"	1/2"	3/8"	10.83	5.71	1.10	1/2"	8.4
Z202016	MF-2016-10	45,402	113,534	3/4"	5/8"	1/2"	10.83	5.71	1.26	5/8"	11.2
Z202220	MF-2220-10	70,528	176,484	7/8"	3/4"	5/8"	10.63	5.51	1.57	3/4"	17.6
Z202622	M-2622-10	90,364	226,169	1"	7/8"	-	13.39	7.09	1.77	-	28.7
Z203226	M-3226-10	125,628	314,299	1-1/4"	1"	-	11.81	7.87	2"	-	32.63
Z203632	M-3632-10	158,688	397,032	-	1-1/4"	-	14.76	8.27	2.36	-	57.3
Z200100	M-100T-10	220,400	551,484	-	-	-	17.72	9.84	2.76	-	94.8
Z200125	M-125T-10*	275,500	689,299	-	-	-	17.72	10.24	3 1/8"	-	125.66

5:1 Design Factor. *Dimension L and E not according to EN 1677-4.



- Made from Quenched & Tempered, Fine Grain Alloy Steel from European Steel Mills.
- Links are individually Proof Tested to values shown, with certification.
- Proof Test with max 60% inside width fixture to prevent localized point loading per ASTM A952.
- Each link is marked with Product Identification Code (PIC) for traceability, Grade, chain size, SWE, GUNNEBO and BG/DGUV manufacturing ID (H32).
- Fatigue rated to at least 20,000 cycles at 1.5 times the Working Load Limit.
- Designed for use with chain, wire or synthetic rope. Applications with wire and synthetic rope generally require a 5:1 Design Factor.
- 3/4-leg requires Gunnebo Industries CG, CGD, CL or CLD components. Engineered Flat compatible with Crosby S-1325 Omega Link.
- Fulfills or exceeds requirements in EN1677:2008, ASTM A952/A952M-02, AS 3775:2014 and AS 3776:2015.

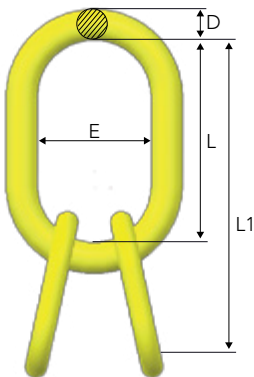
Master Link MFH with engineered flat

Designed for crane hooks, DIN 15401 and 15402. Designed for use with CL, CLD, CG and CGD.

3- and 4-leg chain slings require CLD / CGD.

Stock No.	Code	WLL (lb) 5:1		For chain size			Dimensions (in)				DIN 15401 (mm)	DIN 15402 (mm)	Weight (lb)
		EN 1677-4	A-952/A952M AS 3775.2-2014	1leg	2leg	3-4leg	L	E	D	Flat Thickness			
Z101262	MFH-1310-10	16,530	17,632	1/2"	3/8"	5/16"	9.05	4.92	0.86	3/8"	≤ 12	≤ 16	4.62
Z101263	MFH-1613-10	22,040	29,974	5/8"	1/2"	3/8"	9.84	5.31	1.10	1/2"	≤ 12	≤ 16	8.15
Z101264	MFH-2016-10	37,468	45,402	3/4"	5/8"	1/2"	11.0	5.31	1.25	5/8"	≤ 16	≤ 20	11.6
Z101265	MFH-2220-10	61,712	68,104	7/8"	3/4"	5/8"	12.5	6.88	1.57	3/4"	≤ 25	≤ 32	21.3
Z101266	MFHW-2220-10	61,712	61,712	7/8"	3/4"	5/8"	13.9	8.85	1.57	3/4"	≤ 50	≤ 63	24.4

5:1 Design Factor.



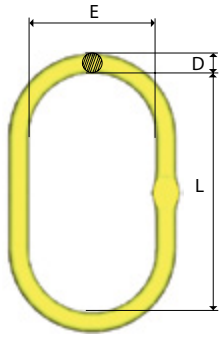
- Made from Quenched & Tempered, Fine Grain Alloy Steel from European Steel Mills.
- Links are individually Proof Tested to values shown, with certification.
- Proof Test with max 60% inside width fixture to prevent localized point loading per ASTM A952.
- Each link is marked with Product Identification Code (PIC) for traceability, Grade, chain size, SWE, GUNNEBO and BG/DGUV manufacturing ID (H32).
- Fatigue rated to at least 20,000 cycles at 1.5 times the Working Load Limit.
- Designed for use with chain, wire or synthetic rope. Applications with wire and synthetic rope generally require a 5:1 Design Factor.
- Engineered Flat on sub links up to MT-16-10.
- Fulfills or exceeds requirements in EN1677:2008, ASTM A952/A952M-02, AS 3775:2014 and AS 3776:2015.

Grade 100 Welded Master Link Assembly MT

Designed for use with chain or wire rope. For 3 and 4-leg slings

Stock No.	Code	WLL (lb)	Proof Load (lb)	Grade 100 Chain Size 3/4-leg (in)	Grade 80 Chain Size 3/4-leg (in)	Dimensions (in)								Weight (lb)
						L1	L	E	D	I	e	d	G*	
Z200600	MT-6-10	9,477	23,606	7/32", 9/32"	7/32", 9/32"	10.2	5.5	3.1	0.7	4.7	2.4	0.5	7/32"	3.5
Z200800	MT-8-10	17,191	42,941	5/16"	5/16", 3/8"	11.8	6.3	3.7	0.9	5.5	3.1	0.7	5/16"	7.3
Z201000	MT-10-10	26,448	66,097	3/8"	1/2"	17.1	10.8	5.7	1.1	6.3	3.7	0.9	3/8"	15.7
Z201300	MT-13-10	46,284	115,782	1/2"	5/8"	18.3	10.8	5.7	1.3	7.5	4.3	1.1	1/2"	24.3
Z201600	MT-16-10	68,324	170,863	5/8"	3/4"	21.5	10.6	5.5	1.6	10.8	5.7	1.3	5/8"	39.7
Z202000	MT-20-10	105,792	264,613	3/4"	7/8"	24.0	13.4	7.1	1.8	10.6	5.5	1.6	3/4"	63.9
Z202200	MT-22-10	132,240	330,935	7/8"	1"	27.2	13.8	7.9	2.2	13.4	7.1	1.8	-	101.4
Z202600	MT-26-10	187,340	468,750	1"	1-1/4"	28.5	14.8	8.3	2.4	13.8	7.9	2.2	-	149.9
Z203200	MT-32-10	275,500	689,299	1-1/4"	-	32.5	17.7	10.2	3.1	14.8	8.3	2.4	-	242.5

5:1 Design Factor. *Thickness of flat on sub link



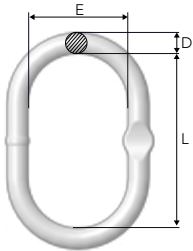
- Engineered flat is an oversized master link designed for use in 1- and 2-leg sling configurations.
- Compatible with CL, CLD, CG, and CGD components for flexible application in lifting assemblies.
- Clearly marked with the product type (M for single-type master link), size designation, and H32 (Gunnepro Industries manufacturing ID) verified by third-party audit from BG in Germany.
- Additional markings include a traceability code and the country of origin to ensure full traceability and identification.
- Meets EN 1677:2008 (WLL +25%), ASTM A952/A952M-02, AS 3775:2014, and AS 3776:2015 standards.
- 5:1 Design Factor

Grade 100 Welded Oversized Master Link MFX

Oversized, for 1- and 2-leg slings. Designed for use with CL, CLD, CG and CGD.

Stock No.	Code	WLL (lb)	Proof Load (lbs)	For Grade 100 Chain Size (in)		Dimensions (in)				Weight (lb)
				For chain 1-leg	For chain 2-leg	L	E	D	Flat Thickness	
Z100923	MFX-108-10 GrabiQ V2	11,905	19,762	3/8"	5/16"	13.4	7.1	7/8"	5/16"	8.1
Z100921	MFX-1310-10 GrabiQ V2	18,078	45,195	1/2"	3/8"	13.4	7.1	1-1/8"	3/8"	10.2
Z100924	MFX-1613-10 GrabiQ V2	29,983	74,957	5/8"	1/2"	13.4	7.1	1-1/4"	1/2"	15.4
Z100922	MFX-2016-10 GrabiQ V2	45,415	113,538	3/4"	5/8"	13.4	7.1	1-9/16"	5/8"	21.3

5:1 Design Factor.



Master Link MF HDG with Engineered Flat

Stock No.	Code	WLL (lb)		Dimensions (in)			Flat Thickness	Weight (lb)
		EN1677-4	A-952/A952M	L	E	D		
BG14489	MF-86-8 HDG	4,500	5,510	4.72	2.36	.51	7/32"	1.10
BG14482	MF-108-8 HDG	7,100	8,800	5.51	3.15	.67	5/16"	1.76
BG14483	MF-1310-8 HDG	12,000	15,000	6.30	3.74	.87	3/8"	3.31
BG14484	MF-1613-8 HDG	18,000	22,600	7.48	4.33	1.10	1/5"	6.17

5:1 Design Factor.



William Hackett
Lifting Products Limited

OFFSHORE



HA Master Links and Quad Assemblies

HA Master Links OS+

The DNV type approved HA range of master links are recognised and specified globally by operators, contractors and rigging shops for use in offshore rigging and DNV 2.7-1 container lifting sets. The industry leading HA master link range benefits from the following:

DNV type approved: certified by DNV pursuant to Type Approval No. TAS00003J2 Rev. 1 and Type Approval No. TAS0000337 Rev. 3 in accordance with DNV-ST-E271, DNV-ST-E273, BSEN ISO 10855-2, EN1677-4, ASME B30.26, and API-2CCU - 2017-08.

Additional standards: In addition to standards covered by DNV type approval, this product is designed and manufactured in accordance with AS3776:2015.

Product sizes: available in diameters from 16mm up to 120mm and working load limits from 4.1 tonnes to 250 tonnes.

Temperature range: - 40°C to + 200°C without reduction in WLL.

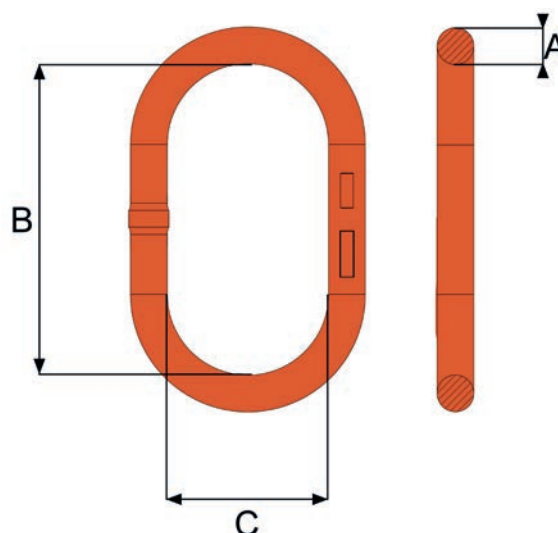
Material & hardness: manufactured from grade 8 triple alloy steel, with a hardness range of 34 to 36 HRC.

Average impact energy (Charpy): 42 Joules minimum impact resistance at - 40°C (up to 50mm diameter).

Testing: individually proof load tested to 2.5 times WLL (for sizes 16mm to 75mm) and 2 times WLL (for sizes 90mm and 120mm), and 100% MPI tested.

Embossing: includes product description (i.e., "HA16ML 8 OS+") and batch number.

Safety factor: 5 : 1.



Metric Specifications

Part Code	Description	A mm	B mm	C mm	WLL t	Mass kg
330.160+	HA16ML 8 OS+	16.0	150	75	4.10	0.68
330.22S+	HA22MLS 8 OS+	22.0	162	90	11.00	1.47
330.220+	HA22ML 8 OS+	22.0	270	140	7.00	2.28
330.250+	HA25ML 8 OS+	25.5	270	140	9.30	3.11
330.28S+	HA28MLS 8 OS+	28.0	200	110	19.50	2.95
330.280+	HA28ML 8 OS+	28.0	270	140	14.50	3.78
330.320+	HA32ML 8 OS+	32.0	270	140	19.00	5.02
330.360+	HA36ML 8 OS+	36.0	270	140	26.00	6.46
330.400+	HA40ML 8 OS+	40.0	280	155	30.50	8.46
330.450+	HA45ML 8 OS+	45.0	320	175	40.00	12.18
330.500+	HA50ML 8 OS+	50.0	350	195	51.00	16.54
330.650+	HA65ML 8 OS+	65.0	410	220	75.00	33.02
330.750+	HA75ML 8 OS+	75.0	450	250	100.00	48.98
330.900	HA90ML*	90.0	510	300	150.00	86.00
330.1200	HA120ML*	120.0	610	410	250.00	197.00

Imperial Specifications

Part Code	Description	A inch	B inch	C inch	WLL lbs	Mass lbs
330.160+	HA16ML 8 OS+	5/8	6	3	9,038	1.50
330.22S+	HA22MLS 8 OS+	7/8	6-1/2	3-5/8	24,250	3.24
330.220+	HA22ML 8 OS+	7/8	10-3/4	5-5/8	15,432	5.03
330.250+	HA25ML 8 OS+	1	10-3/4	5-5/8	20,503	6.86
330.28S+	HA28MLS 8 OS+	1-1/8	8	4-3/8	42,990	6.50
330.280+	HA28ML 8 OS+	1-1/8	10-3/4	5-5/8	31,967	8.33
330.320+	HA32ML 8 OS+	1-1/4	10-3/4	5-5/8	41,887	11.07
330.360+	HA36ML 8 OS+	1-7/16	10-3/4	5-5/8	57,320	14.29
330.400+	HA40ML 8 OS+	1-9/16	11	6	67,240	18.65
330.450+	HA45ML 8 OS+	1-3/4	12-3/4	7	88,184	26.85
330.500+	HA50ML 8 OS+	2	14	7-3/4	112,436	36.46
330.650+	HA65ML 8 OS+	2-9/16	16-1/8	8-5/8	165,347	72.80
330.750+	HA75ML 8 OS+	3	18	10	220,462	108.00
330.900*	HA90ML*	3-5/8	21	12-1/2	330,693	189.59
330.1200*	HA120ML*	4-3/4	24	16-3/8	551,156	434.31

*Certified on a batch basis to the requirements of DNV 2.7-1

HA Master Links and Quad Assemblies

HA Quad Assemblies OS+

The DNV type approved HA range of quad assemblies are recognised and specified globally by operators, contractors and rigging shops for use in offshore rigging and DNV 2.7-1 container lifting sets. The industry leading HA quad assembly range benefits from the following:

DNV type approved: certified by DNV pursuant to Type Approval No. TAS00003J2 Rev. 1 and Type Approval No. TAS0000337 Rev. 3 in accordance with DNV-ST-E271, DNV-ST-E273, BSEN ISO 10855-2, EN1677-4, ASME B30.26, and API-2CCU - 2017-08.

Additional standards: In addition to standards covered by DNV type approval, this product is designed and manufactured in accordance with AS3776:2015.

Product sizes: available in diameters from 16mm up to 90mm and working load limits from 4.1 tonnes to 150 tonnes.

Temperature range: - 40°C to + 200°C without reduction in WLL.

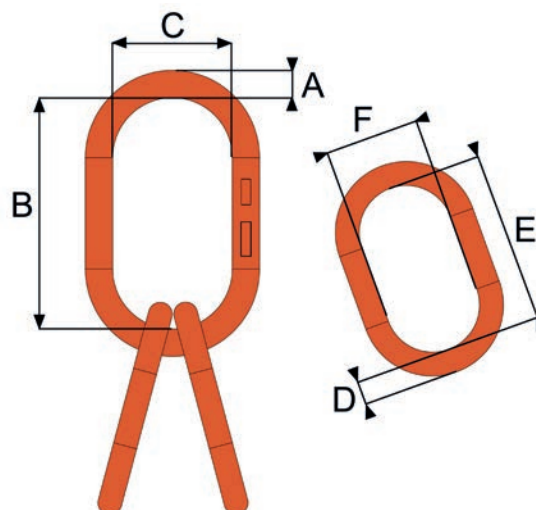
Material & hardness: manufactured from grade 8 triple alloy steel, with a hardness range of 34 to 36 HRC.

Average impact energy (Charpy): 42 Joules minimum impact resistance at - 40°C (up to 50mm diameter).

Testing: individually proof load tested to 2.5 times WLL (for sizes 16mm to 75mm) and 2 times WLL (for size 90mm), and 100% MPI tested.

Embossing: includes product description (i.e., "HA16QA 8 OS+") and batch number.

Safety factor: 5 : 1.



Metric Specifications

Part Code	Description	A mm	B mm	C mm	D mm	E mm	F mm	WLL t	Mass kg
350.160+	HA16QA 8 OS+	16.0	150	75	14.5	125	60	4.10	1.62
350.220+	HA22QA 8 OS+	22.0	162	90	22.0	162	90	11.00	4.41
350.230+	HA23QA 8 OS+	22.0	270	140	22.0	162	90	7.00	5.22
350.250+	HA25QA 8 OS+	25.5	270	140	22.0	162	90	9.30	6.05
350.260+	HA26QA 8 OS+	28.0	270	140	22.0	162	90	14.50	6.73
350.320+	HA32QA 8 OS+	32.0	270	140	28.0	200	110	19.00	10.92
350.360+	HA36QA 8 OS+	36.0	270	140	28.0	200	110	26.00	12.35
350.400+	HA40QA 8 OS+	40.0	280	155	32.0	270	140	30.50	18.50
350.450+	HA45QA 8 OS+	45.0	320	175	36.0	270	140	40.00	25.09
350.500+	HA50QA 8 OS+	50.0	350	195	45.0	320	175	51.00	40.89
350.650+	HA65QA 8 OS+	65.0	410	220	50.0	350	195	75.00	66.10
350.750+	HA75QA 8 OS+	75.0	450	250	65.0	410	220	100.00	115.02
350.900	HA90QA*	90.0	510	300	70.0	400	200	150.00	164.10

Imperial Specifications

Part Code	Description	A inch	B inch	C inch	D inch	E inch	F inch	WLL lbs	Mass lbs
350.160+	HA16QA 8 OS+	5/8	6	3	9/16	5	2-3/8	9,038	3.57
350.220+	HA22QA 8 OS+	7/8	6-1/2	3-5/8	7/8	6-3/8	3-9/16	24,250	9.72
350.230+	HA23QA 8 OS+	7/8	10-3/4	5-5/8	7/8	6-3/8	3-9/16	15,432	11.50
350.250+	HA25QA 8 OS+	1	10-3/4	5-5/8	7/8	6-3/8	3-9/16	20,502	13.33
350.260+	HA26QA 8 OS+	1-1/8	10-3/4	5-5/8	7/8	6-3/8	3-9/16	31,966	14.83
350.320+	HA32QA 8 OS+	1-1/4	10-3/4	5-5/8	1-1/8	7-7/8	4-5/16	41,887	24.07
350.360+	HA36QA 8 OS+	1-7/16	10-3/4	5-5/8	1-1/8	7-7/8	4-5/16	57,319	27.22
350.400+	HA40QA 8 OS+	1-9/16	11	6	1-1/4	10-5/8	5-1/2	67,240	40.78
350.450+	HA45QA 8 OS+	1-3/4	12-3/4	7	1-7/16	10-5/8	5-1/2	88,184	55.31
350.500+	HA50QA 8 OS+	2	14	7-3/4	1-3/4	12-5/8	6-7/8	112,434	90.14
350.650+	HA65QA 8 OS+	2-9/16	16-1/8	8-5/8	2	13-3/4	7-11/16	165,345	145.72
350.750+	HA75QA 8 OS+	3	18	10	2-9/16	16-1/8	8-11/16	220,460	253.57
350.900	HA90QA*	3-5/8	21	12-1/2	2-3/4	15-3/4	7-7/8	330,600	360.00

*Certified on a batch basis to the requirements of DNV 2.7-1

Zinc-Tough® Technology

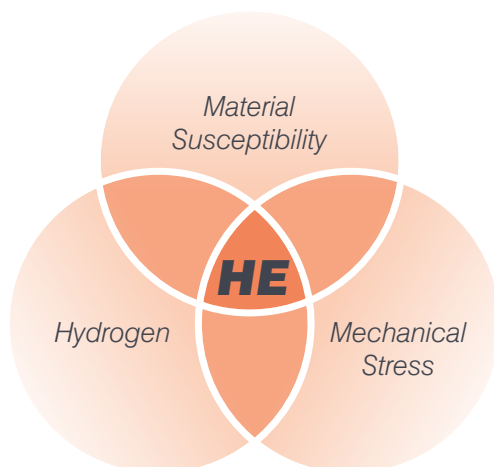
William Hackett and McKinnon Chain have combined their technical knowledge and worked alongside a wide range of stakeholders, to develop Zinc-Tough® Technology. The Zinc-Tough® range of chain and master links supplied by William Hackett provides superior corrosion protection whilst complementing the steel's inherent resistance to hydrogen embrittlement (HE) and hydrogen assisted stress cracking (HASC), making them ideal for use in both marine and aquaculture applications within an offshore environment.

What is hydrogen embrittlement (HE)?

Hydrogen embrittlement (HE) is the process by which metals such as steel become brittle and fracture under stress, this is due to the introduction and subsequent diffusion of hydrogen into the metal.

Root causes of hydrogen embrittlement (HE)

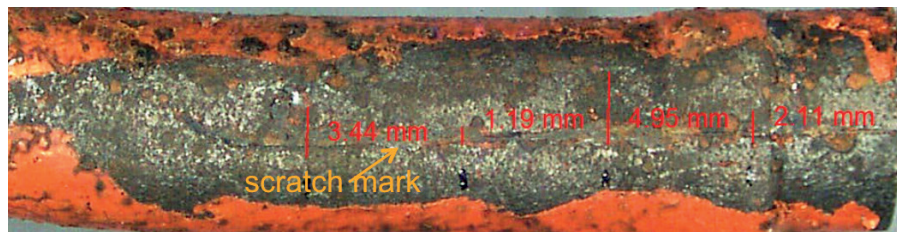
For hydrogen embrittlement (HE) to take place, three elements are required: a source of hydrogen, material susceptibility (or hardness of the steel) and mechanical stresses.



Rust penetration below the powder coating at 600 hours

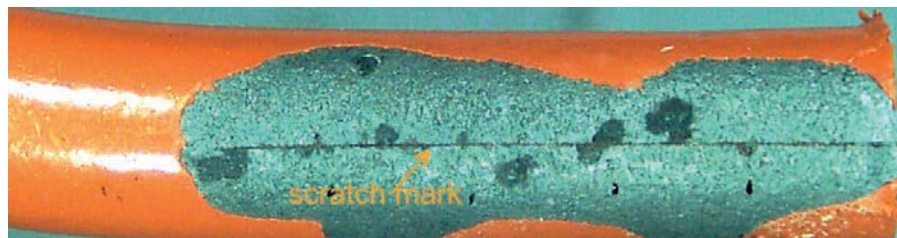
1. Standard link with powder coated surface only

Corrosion creeps beneath the powder coated layer. The powder coating flakes off easily.



2. Zinc-Tough® link with zinc thermal diffusion treatment and powder coating

No corrosion creepage. The powder coat layer remains intact.



Benefits of ZINC-TOUGH® TECHNOLOGY



Non interference with metallurgical properties

The low temperature application of the coating process has no interference with hardness and toughness properties



Zero risk to hydrogen embrittlement

Many coating processes such as galvanizing and electroplating bring with them the risk of embrittlement. This process poses a zero risk of hydrogen embrittlement.



Corrosion protection

The zinc-iron alloy coating is a sacrificial coating providing galvanic protection.



Excellent bonding surface and spark free

The zinc layers are spark free and provide an excellent bonding surface and is ideal for powder coating.



Abrasion resistant

The coating approximately 45µ in thickness is highly abrasion resistant.



Environmentally friendly process

Zinc thermal diffusion is an environmental friendly process (a non-toxic, heavy metal free process with minimum waste).

HA Master Links and Quad Assemblies - Zinc-Tough®

HA Master Links OS+ Zinc-Tough®

The DNV type approved HA range of master links are available with the additional benefit of Zinc-Tough® technology. This zinc thermal diffusion surface treatment provides superior protection against hydrogen embrittlement and stress corrosion cracking. This makes the Zinc-Tough® HA master link range the optimal solution for lifting operations in the harshest offshore environments. Alongside Zinc-Tough®, our HA master link range benefits from the following:

DNV type approved: certified by DNV pursuant to Type Approval No. TAS00003J4 Rev. 1 and Type Approval No. TAS0000333 Rev. 3 in accordance with DNV-ST-E271, DNV-ST-E273, BSEN ISO 10855-2, EN1677-4, ASME B30.26, and API-2CCU - 2017-08.

Additional standards: In addition to standards covered by DNV type approval, this product is designed and manufactured in accordance with AS3776:2015.

Product sizes: available in diameters from 22mm up to 75mm and working load limits from 7 tonnes to 100 tonnes.

Temperature range: - 40°C to + 200°C without reduction in WLL.

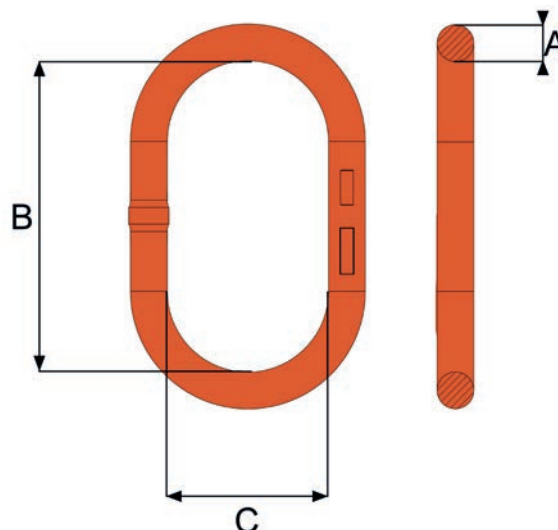
Material & hardness: manufactured from grade 8 triple alloy steel with Zinc-Tough® treatment, and a hardness range of 34 to 36 HRC.

Average impact energy (Charpy): 42 Joules minimum impact resistance at - 40°C (up to 50mm diameter).

Testing: individually proof load tested to 2.5 times working load limit, and 100% MPI tested.

Embossing: includes product description (i.e., "HA22ML OS+ ZT") and batch number.

Safety factor: 5 : 1.



Metric Specifications

Part Code	Description	A mm	B mm	C mm	WLL t	Mass kg
330.220.+ZT	HA22ML OS+ ZT	22.0	270	140	7.00	2.28
330.250.+ZT	HA25ML OS+ ZT	25.5	270	140	9.30	3.11
330.280.+ZT	HA28ML OS+ ZT	28.0	270	140	14.50	3.78
330.320.+ZT	HA32ML OS+ ZT	32.0	270	140	19.00	5.02
330.360.+ZT	HA36ML OS+ ZT	36.0	270	140	26.00	6.46
330.400.+ZT	HA40ML OS+ ZT	40.0	280	155	30.50	8.46
330.450.+ZT	HA45ML OS+ ZT	45.0	320	175	40.00	12.18
330.500.+ZT	HA50ML OS+ ZT	50.0	350	195	51.00	16.54
330.650.+ZT	HA65ML OS+ ZT	65.0	410	220	75.00	33.02
330.750.+ZT	HA75ML OS+ ZT	75.0	450	250	100.00	48.98

Imperial Specifications

Part Code	Description	A inch	B inch	C inch	WLL lbs	Mass lbs
330.220.+ZT	HA22ML OS+ ZT	7/8	10-3/4	5-5/8	15,432	5.03
330.250.+ZT	HA25ML OS+ ZT	1	10-3/4	5-5/8	20,503	6.86
330.280.+ZT	HA28ML OS+ ZT	1-1/8	10-3/4	5-5/8	31,967	8.33
330.320.+ZT	HA32ML OS+ ZT	1-1/4	10-3/4	5-5/8	41,887	11.07
330.360.+ZT	HA36ML OS+ ZT	1-7/16	10-3/4	5-5/8	57,320	14.29
330.400.+ZT	HA40ML OS+ ZT	1-9/16	11	6	67,240	18.65
330.450.+ZT	HA45ML OS+ ZT	1-3/4	12-3/4	7	88,184	26.85
330.500.+ZT	HA50ML OS+ ZT	2	14"	7-3/4	112,436	36.46
330.650.+ZT	HA65ML OS+ ZT	2-9/16	16-1/8	8-5/8	165,347	72.80
330.750.+ZT	HA75ML OS+ ZT	3	18	10	220,462	108.00



HA Master Links and Quad Assemblies - Zinc-Tough®

HA Quad Assemblies OS+ Zinc-Tough®

The DNV type approved HA range of quad assemblies are available with the additional benefit of Zinc-Tough® technology. This zinc thermal diffusion surface treatment, provides superior protection against hydrogen embrittlement and stress corrosion cracking. This makes Zinc-Tough® HA quad assembly range the optimal solution for lifting operations in the harshest offshore environments. Alongside Zinc-Tough®, our HA quad assembly range benefits from the following:

DNV type approved: certified by DNV pursuant to Type Approval No. TAS00003J4 Rev. 1 and Type Approval No. TAS0000333 Rev. 3 in accordance with DNV-ST-E271, DNV-ST-E273, BSEN ISO 10855-2, EN1677-4, ASME B30.26, and API-2CCU - 2017-08.

Additional standards: In addition to standards covered by DNV type approval, this product is designed and manufactured in accordance with AS3776:2015.

Product sizes: available in diameters from 22mm up to 75mm and working load limits from 7 tonnes to 100 tonnes.

Temperature range: - 40°C to + 200°C without reduction in WLL.

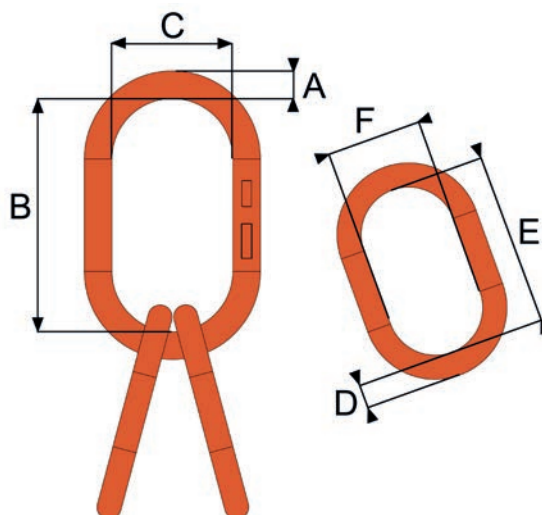
Material & hardness: manufactured from grade 8 triple alloy steel with Zinc-Tough® treatment, and a hardness range of 34 to 36 HRC.

Average impact energy (Charpy): 42 Joules minimum impact resistance at - 40°C (up to 50mm diameter).

Testing: individually proof load tested to 2.5 times working load limit, and 100% MPI tested.

Embossing: includes product description (i.e., "HA25QA OS+ ZT") and batch number.

Safety factor: 5 : 1.



Metric Specifications

Part Code	Description	A mm	B mm	C mm	D mm	E mm	F mm	WLL t	Mass kg
350.230.+ZT	HA23QA OS+ ZT	22.0	270	140	22.0	162	90	7.00	5.22
350.250.+ZT	HA25QA OS+ ZT	25.5	270	140	22.0	162	90	9.30	6.05
350.260.+ZT	HA26QA OS+ ZT	28.0	270	140	22.0	162	90	14.50	6.73
350.320.+ZT	HA32QA OS+ ZT	32.0	270	140	28.0	200	110	19.00	10.92
350.360.+ZT	HA36QA OS+ ZT	36.0	270	140	28.0	200	110	26.00	12.35
350.400.+ZT	HA40QA OS+ ZT	40.0	280	155	32.0	270	140	30.50	18.50
350.450.+ZT	HA45QA OS+ ZT	45.0	320	175	36.0	270	140	40.00	25.09
350.500.+ZT	HA50QA OS+ ZT	50.0	350	195	45.0	320	175	51.00	40.89
350.650.+ZT	HA65QA OS+ ZT	65.0	410	220	50.0	350	195	75.00	66.10
350.750.+ZT	HA75QA OS+ ZT	75.0	450	250	65.0	410	220	100.00	115.02

Imperial Specifications

Part Code	Description	A inch	B inch	C inch	D inch	E inch	F inch	WLL lbs	Mass lbs
350.230.+ZT	HA23QA OS+ ZT	7/8	10-3/4	5-5/8	7/8	6-3/8	3-9/16	15,432	11.50
350.250.+ZT	HA25QA OS+ ZT	1	10-3/4	5-5/8	7/8	6-3/8	3-9/16	20,502	13.33
350.260.+ZT	HA26QA OS+ ZT	1-1/8	10-3/4	5-5/8	7/8	6-3/8	3-9/16	31,966	14.83
350.320.+ZT	HA32QA OS+ ZT	1-1/4	10-3/4	5-5/8	1-1/8	7-7/8	4-5/16	41,887	24.07
350.360.+ZT	HA36QA OS+ ZT	1-7/16	10-3/4	5-5/8	1-1/8	7-7/8	4-5/16	57,319	27.22
350.400.+ZT	HA40QA OS+ ZT	1-9/16	11	6	1-1/4	10-5/8	5-1/2	67,240	40.78
350.450.+ZT	HA45QA OS+ ZT	1-3/4	12-3/4	7	1-7/16	10-5/8	5-1/2	88,184	55.31
350.500.+ZT	HA50QA OS+ ZT	2	14	7-3/4	1-3/4	12-5/8	6-7/8	112,434	90.14
350.650.+ZT	HA65QA OS+ ZT	2-9/16	16-1/8	8-5/8	2	13-3/4	7-11/16	165,345	145.72
350.750.+ZT	HA75QA OS+ ZT	3	18	10	2-9/16	16-1/8	8-11/16	220,460	253.57



HA Master Links with Handles

HA Master Links with Handles

The HA range of master links with handles are designed specifically for the lifting and lowering of lifeboats. All HA master links with handles have a primary application of Zinc-Tough® corrosion protection treatment, complimented by a secondary application of orange powder coating. These links also benefit from the following:

Standards: designed and manufactured in accordance with DNV-ST-E271, EN1677-4, ASME B30.26, and AS3776-2015.

Product sizes: available in diameters from 22mm up to 36mm and working load limits from 7 tonnes to 26 tonnes.

Temperature range: - 40°C to + 200°C without reduction in WLL.

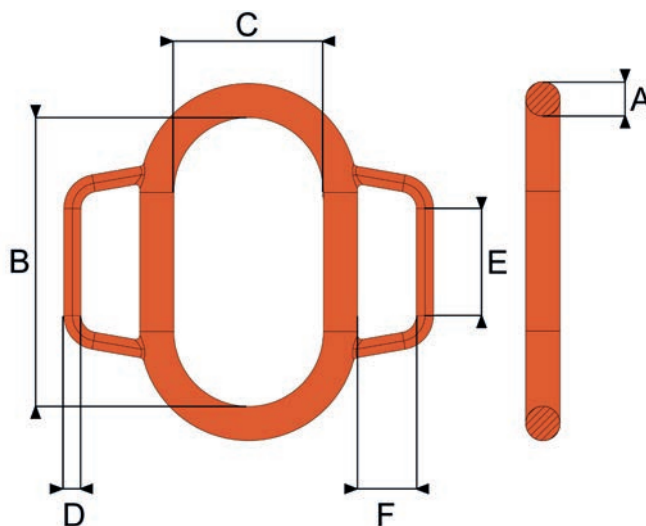
Material & hardness: manufactured from grade 8 triple alloy steel with Zinc-Tough® treatment, and a hardness range of 34 to 36 HRC.

Average impact energy (Charpy): 42 Joules minimum impact resistance at - 40°C.

Testing: individually proof load tested to 2.5 times working load limit, and 100% MPI tested.

Embossing: includes product description (i.e., "HA-22 H8 OS+") and batch number.

Safety factor: 5 : 1.



Metric Specifications

Part Code	Description	A mm	B mm	C mm	D mm	E mm	F mm	WLL t	Mass kg
330.220.H	HA22 H8 OS+	22.0	270	140	16	100	55	7.00	3.00
330.250.H	HA25 H8 OS+	25.5	270	140	16	100	55	9.30	3.84
330.280.H	HA28 H8 OS+	28.0	270	140	16	100	55	14.50	4.51
330.320.H	HA32 H8 OS+	32.0	270	140	16	100	55	19.00	5.75
330.360.H	HA36 H8 OS+	36.0	270	140	16	100	55	26.00	7.19

Imperial Specifications

Part Code	Description	A inch	B inch	C inch	D inch	E inch	F inch	WLL lbs	Mass lbs
330.220.H	HA22 H8 OS+	7/8	10-5/8	5-1/2	5/8	4	2-1/8	15,432	6.61
330.250.H	HA25 H8 OS+	1	10-5/8	5-1/2	5/8	4	2-1/8	20,503	8.46
330.280.H	HA28 H8 OS+	1-1/8	10-5/8	5-1/2	5/8	4	2-1/8	31,967	9.94
330.320.H	HA32 H8 OS+	1-1/4	10-5/8	5-1/2	5/8	4	2-1/8	41,888	12.67
330.360.H	HA36 H8 OS+	1-1/2	10-5/8	5-1/2	5/8	4	2-1/8	57,320	15.85

LIFTING SLING FITTINGS FOR WIRE ROPE

MASTER LINKS & MASTER LINK ASSEMBLIES



Applications

Master links are the connection between the (crane) hook and wire rope fittings. The Green Pin® wire rope master links have a large opening to accommodate for larger lifting points and crane hooks. The welded master link designed for 1 or 2 legs and the master link assembly for 3 or 4 legs. A flat part on the link enables assembly with an omega link or with a thimble.

More information about the range, certification and finish, please go to the introduction of chapter 3 lifting slings.

Design

The Green Pin® Wider Master Link and Master Link Assembly are Grade 100 master links with larger inside dimensions.

The Green Pin® DNV Master Link and Master Link Assembly are grade 80 master links and are perfect for offshore use. This welded master link is DNV-certified.

The Green Pin® Heavy Duty Master Link is a grade 80 master link for heavy-duty applications. The consistent Charpy V value of at least 42 J at -4 °F and 27 J at -40 °F throughout the entire master link, makes it compliant with offshore certification requirements. The master links are fully forged as a single piece.

Wire rope master links are generally marked with:

- manufacturer's symbol - e.g. GP
- size in mm and/ or inch - e.g. 25 and/or 1"
- traceability code - e.g. HA
- steel grade (specific products) - e.g. 8 or 10
- item code (specific products) - e.g. UMSW

As prescribed by the Machinery Directive 2006/42/EC our hooks, master links and connecting links are supplied with a CE Declaration of Incorporation (CE IIB), stating the products are in compliance with the requirements of the machinery directive. These products do not have a CE marking, since CE markings are only to be provided for machines, whereas hooks, master links and connecting links are components of machines, and not machines themselves. Providing components with a CE marking is an incorrect interpretation of the Machinery Directive.

Machinery definition according to Machinery Directive 2006/42/EC – Article 2 – definitions (a) and Machinery Regulation 2023/1230/EU

“An assembly of linked parts or components, at least one of which moves and which are joined together, intended for lifting loads and whose only power source is directly applied human effort.”

“The maximum working load shall be prominently marked on the lifting machinery. The marking shall be legible, indelible and in an un-coded form. Where the maximum working load depends on the configuration of the lifting machinery, each operating position shall be provided with a load plate indicating, preferably in diagrammatic form or by means of tables, the working load permitted for each configuration.”

This definition means that a sling is a Machine and that the hooks, master links and connecting links are not considered as a machine themselves, but as components of a machine. The CE and the WLL markings in consequence, are mandatory only on the assembled sling tag as the sling is the machine according to MD 2006/42/EC. This rule is still valid with the new Machine Regulation 2023/1230 which will replace the MD cited previously within early 2027.

Instructions for use

Master links and master link assemblies, should be inspected before use to ensure that:

- all markings are legible.
- Master links and master link assemblies with the correct WLL has been selected with respect to the sling design.
- Master links and master link assemblies are free from nicks, gouges, and cracks.
- Master links and master link assemblies are not heat treated, modified, repaired, or reshaped by machining or bended. (This may affect their Working Load Limit).
- Master links and master link assemblies are not distorted or unduly worn.

Master links and master link assemblies must be regularly inspected in accordance with the safety standards given in the country of use. This is required because the products in use may be affected by issues such as wear, misuse and overloading, which may lead to deformation and alteration of the material structure. Inspection should take place at least every six months, and more frequently when the links are used in severe operating conditions.

Symmetry of loading

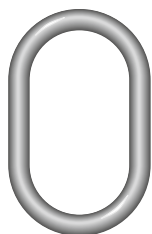
The WLL values mentioned are based on symmetrical loading of the sling. This means that when the load is lifted the sling legs are symmetrically distributed in the plane and all legs of the sling have the same angle to the vertical. The loading can be assumed to be symmetric if all the following conditions are met:

- the load is less than 80 % of marked WLL, and
- sling leg angles to the horizontal are all less than 75° and
- sling leg angles to the vertical are all within 15° to each other, and
- in the case of three- and four-leg slings, the plane angles are within 15° of each other.

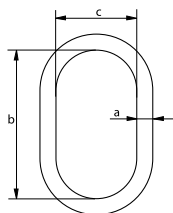
If one of the above parameters is not met, the loading should be considered to be asymmetric, and the lift should be referred to a competent engineer to establish the safe rating for the sling. Alternatively, in the case of asymmetric loading, the sling should be derated to half the marked WLL. If the load tends to tilt during the lift, it should be lowered, and the attachments changed by repositioning the attachment points or by using compatible shortening devices. The safety factor of 4 or 5 on the individual components is designed for safety only. Never exceed the indicated WLL.

Use the below reduction table if a multi leg sling for wire rope fittings is not used for the purpose for which it has been designed, for example for a lifting operation with fewer legs than the number of legs of the sling:

Types of chain sling	Number of legs used	Factor to apply to marked WLL
Two-leg	1	1/2
Three- and four-leg	2	2/3
Three- and four-leg	1	1/3



P-6810



Green Pin® DNV Master Link

Grade 80 master link

- **Material:** alloy steel, grade 80
- **Safety factor:** MBL equals 5 x WLL
- **Standard:** DNV 2.7-1, EN 12079-2 and EN 1677-4
- **Finish:** painted silver
- **Temperature range:** -40 °F up to +392 °F
- **Certification:** 2.1 2.2 3.1 MTC[®] MPI[®] DNV 2.7-1^a CE II^B
- **Article code:** scan QR code to see article codes
- **Note:** size WLL 75 t and 100 t have a temperature range of -4 °F up to +392 °F

Scan for additional product details



working load limit	diameter	length inside	width inside	weight each
metric t	a inch	b inch	c inch	lbs
4.1	$\frac{5}{8}$	$5 \frac{29}{32}$	$2 \frac{15}{16}$	1.5
7	$\frac{7}{8}$	$10 \frac{5}{8}$	$5 \frac{1}{2}$	5.03
9.3	$1 \frac{1}{32}$	$10 \frac{5}{8}$	$5 \frac{1}{2}$	6.86
14.5	$1 \frac{3}{32}$	$10 \frac{5}{8}$	$5 \frac{1}{2}$	8.33
19.5	$1 \frac{3}{32}$	$7 \frac{7}{8}$	$4 \frac{11}{32}$	6.61
19	$1 \frac{1}{4}$	$10 \frac{5}{8}$	$5 \frac{1}{2}$	11.07
26	$1 \frac{13}{32}$	$10 \frac{5}{8}$	$5 \frac{1}{2}$	14.24
30.5	$1 \frac{9}{16}$	$11 \frac{1}{32}$	$6 \frac{3}{32}$	18.65
45	$1 \frac{25}{32}$	$12 \frac{19}{32}$	$6 \frac{7}{8}$	26.9
51	$1 \frac{31}{32}$	$13 \frac{25}{32}$	$7 \frac{11}{16}$	36.4
75	$2 \frac{9}{16}$	$16 \frac{5}{32}$	$8 \frac{21}{32}$	73
100	$2 \frac{15}{16}$	$17 \frac{23}{32}$	$9 \frac{27}{32}$	108

CAD INFO



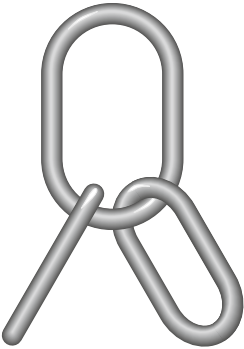


Green Pin® DNV Master Link Assembly

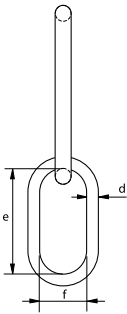
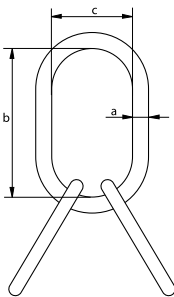
Grade 80 master link assembly

- **Material:** alloy steel, grade 80
- **Safety factor:** MBL equals 5 x WLL
- **Standard:** DNV 2.7-1, EN 12079-2 and EN 1677-4
- **Finish:** painted silver
- **Temperature range:** -40 °F up to +392 °F
- **Certification:** 2.1 2.2 3.1 MTC^b MPI^b DNV 2.7-1^a CE II B
- **Article code:** scan QR code to see article codes
- **Note:** size WLL 75 t and 100 t have a temperature range of -4 °F up to +392 °F

Scan for additional product details



P-6820



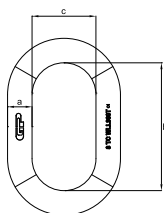
working load limit	diameter	length inside	width inside	diameter	length inside	width inside	weight each
metric t	a inch	b inch	c inch	d inch	e inch	f inch	lbs
4.1	5/8	5 29/32	2 15/16	9/16	4 29/32	2 3/8	3.57
7	7/8	10 5/8	5 1/2	7/8	6 3/8	3 17/32	11.51
9.3	1 1/32	10 5/8	5 1/2	7/8	6 3/8	3 17/32	13.34
14.5	1 3/32	10 5/8	5 1/2	7/8	6 3/8	3 17/32	14.84
19	1 1/4	10 5/8	5 1/2	1 3/32	7 7/8	4 11/32	24
26	1 13/32	10 5/8	5 1/2	1 3/32	7 7/8	4 11/32	27.3
30.5	1 9/16	11 1/32	6 3/32	1 1/4	10 5/8	5 1/2	40.8
40	1 25/32	12 19/32	6 7/8	1 13/32	10 5/8	5 1/2	55
51	1 31/32	13 25/32	7 11/16	1 25/32	12 19/32	6 7/8	90
75	2 9/16	16 5/32	8 21/32	1 31/32	13 25/32	7 11/16	146
100	2 15/16	17 23/32	9 27/32	2 9/16	16 5/32	8 21/32	254

CAD INFO





P-6815



Green Pin® Heavy Duty Master Link

Grade 80 weldless heavy duty master link

- **Material:** alloy steel, grade 80
- **Safety factor:** MBL equals 5 x WLL
- **Standard:** ASME B30.26
- **Finish:** painted silver and green
- **Temperature range:** -40 °F up to +392 °F
- **Certification:** 2.1 2.2 3.1 MTC^a MPI^b US^b CE IIA
- **Article code:** scan QR code to see article codes



working load limit	diameter	length inside	width inside	weight each
metric t	a inch	b inch	c inch	lbs
131	3 ¹⁷ / ₃₂	18 ¹ / ₈	11 ¹³ / ₁₆	170
157	3 ¹⁵ / ₁₆	19 ¹¹ / ₁₆	11 ¹³ / ₁₆	225
250	4 ¹⁷ / ₃₂	23 ⁵ / ₈	15 ³ / ₄	362
300	4 ¹⁷ / ₃₂	23 ⁵ / ₈	11 ¹³ / ₁₆	342
400	4 ¹⁷ / ₃₂	19 ⁹ / ₃₂	9 ²⁷ / ₃₂	291

CAD



CM[®] **CHAIN & RIGGING ATTACHMENTS**

SHACKLES

**CHAIN & SYNTHETIC
SLING COMPONENTS**

RIGGING HARDWARE

CLAMPS

**TRANSPORTATION
CHAIN & ATTACHMENTS**

SPECIALTY PRODUCTS



C O L U M B U S M c K I N N O N C O R P O R A T I O N



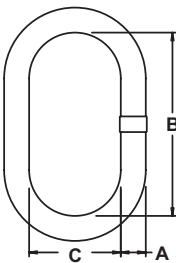
MASTER LINK DUAL RATED FOR USE WITH HA800 OR HA1000

WORKING LOAD LIMIT: 5,400 TO 187,800 LBS.

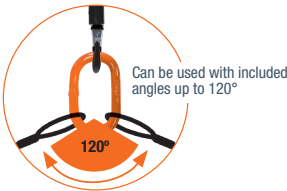


BENEFITS & FEATURES

- Accepts both Herc-Alloy[®] 800 & 1000 chain and components
- Durable orange powder coated finish
- May be used for mechanical and welded sling assemblies
- 100% proof tested
- Design factor 4:1



Trade Size (in.)	Working Load Limit (lbs.)	Product Code	Nominal Dimensions (in.)			Weight (lbs.)	Type and Size of Grade 80 Chain Sling (in.)	
			Material Diameter A	Inside Length B	Inside Width C		Single	Double
Standard Sizes (DUAL RATED THROUGH 1-3/4" SIZE ONLY)								
3/8	5,400	555231	0.41	3.00	1.50	0.33	7/32 & 9/32	7/32
1/2	8,600	555232	0.56	5.00	2.50	1.02	3/8	9/32
3/4	17,600	555235	0.75	5.50	2.75	2.08	1/2	3/8
1	30,000	555238	1.00	7.00	3.50	4.59	5/8 & 3/4	1/2
1-1/4	45,200	555240	1.25	8.75	4.38	9.31	7/8	5/8
1-1/2	70,600	555243	1.50	10.50	5.25	15.60	1	3/4 & 7/8
1-3/4	105,900	555246	1.75	12.00	6.00	24.40	1-1/4	1
2*	88,900	554949	2.00	14.00	7.00	38.00	1-1/4	1
2-1/4*	125,200	554951	2.25	16.00	8.00	55.00	—	1-1/4
2-3/4*	187,800	554957	2.75	16.00	9.00	84.84	—	—
Special Sizes (NOT DUAL RATED)								
1	18,200	554969	1.00	8.00	4.00	5.25	N/A	N/A
1-1/4	22,800	554942	1.25	12.00	6.00	12.08		
1-1/2	36,700	554944	1.50	14.00	7.00	20.16		
1-1/2	42,900	554970	1.50	12.00	6.00	17.60		
1-3/4	57,100	554945	1.75	14.00	7.00	27.92		
2	77,800	554952	2.00	16.00	8.00	43.00		



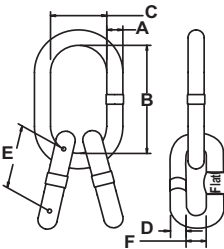
NOTE: Standard sizes dual rated through 1-3/4" only. Special sizes not dual rated.

WIDE BODY SUB-ASSEMBLY WITH FLATS HERC-ALLOY 800®

WORKING LOAD LIMIT: 4,600 TO 178,050 LBS.

BENEFITS & FEATURES

- Designed to accept Herc-Alloy 800® chain, wire rope and synthetic attachments
- Durable orange powder coated finish
- 100% proof tested
- May be used for mechanical and welded sling assemblies
- Extra wide body is ideal for wire rope applications
- Sizes up to 1-1/4" intermediate links available with flats to accommodate Omega links
- Design factor 6:1

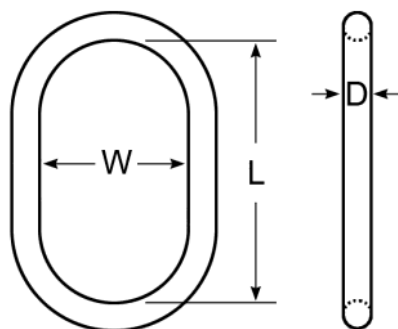


Can be used with included angles up to 120°

Trade Size (in.)	Working Load Limit (lbs.)		Master Link Nominal Dimensions (in.)							Weight (lbs.)	Type and Size of Chain Sling (in.)	
	With Chain	With Wire Rope & Synthetic	Product Code	Material Diameter A	Inside Length B	Inside Width C	Material Diameter D	Inside Length E	Inside Width F		Triple	Quad
1/2	5,750	4,600	ML050SA	0.56	4.84	2.69	0.44	4.13	2.29	1.8	—	—
5/8	9,000	7,200	ML063SA	0.63	5.29	2.98	0.56	4.84	2.69	3.63	7/32	7/32
3/4	14,000	11,200	ML075SA	0.75	6.61	3.72	0.63	5.29	2.98	5.58	9/32	9/32
7/8	17,300	13,840	ML087SA	0.88	7.35	4.14	0.63	5.29	2.98	6.66	—	—
1	26,500	21,200	ML100SA	1.00	7.53	4.30	0.75	6.61	3.72	10.2	3/8	3/8
1-1/4	37,400	29,920	ML125SA	1.25	9.26	5.29	1.00	7.53	4.30	20.0	1/2	1/2
1-1/2	53,000	42,400	ML150SA	1.50	11.03	6.30	1.00	7.53	4.30	26.6	—	—
1-3/4	72,150	57,720	ML175SA	1.75	12.86	7.35	1.25	9.26	5.29	44.3	5/8	5/8
2	94,200	75,360	ML200SA	2.00	14.70	8.40	1.50	11.03	6.30	73.4	3/4	3/4
2-1/4	119,200	95,360	ML225SA	2.25	16.54	9.45	1.75	12.86	7.35	108.2	7/8	7/8

NOTE: Master link with flats are available through 1-1/4" only

OBLONG MASTER LINKS



OBLONG MASTER LINKS, ALLOY, DOMESTIC

Diameter in Inches	DOMESTIC W.L.L. in Pounds*	Link Dimensions in Inches (+/- 4%)		Approximate Weight Each in Pounds
		-W-	-L-	
		Width	Length	
5/8	9,000	3.0	6.0	1.50
1.1/2	54,300	5.1/4	10.5	16.50
1.3/4	84,900	6.0	12.0	24.80
2	102,600	7.0	14.0	37.60
2.1/4	143,100	8.0	16.0	54.30
2.1/2	160,000	8.0	16.0	68.00
2.3/4	216,900	9.0	16.0	85.60
3	228,000	10.0	19.0	136.00
4	373,000	10.0	20.0	228.00
4.1/2	360,000	14.0	28.0	345.00
5	395,000	15.0	30.0	516.00
5.1/2	495,900	16.0	32.0	675.00
6	606,100	18.0	36.0	875.00

A.S.T.M. A952/A906*. Welded or forged*, painted.

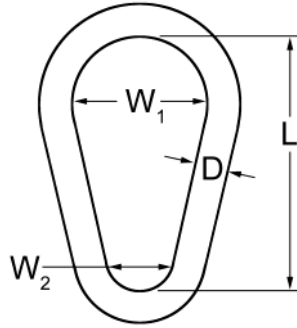
ALL oblong master links are individually proof tested in accordance with the requirements of ASTM A952 / A906.

All oblong master links meet or exceed the performance requirements of A.S.T.M. A952 - Standard Specification for Forged Grade 80 Alloy Steel Lifting Components and Welded Attachment Links and the mechanical requirements of A.S.T.M. 906, Standard Specification for Alloy Steel Chain Slings for Overhead Lifting - as applicable.

* All welded master links have a 4:1 design factor — Forged (weldless) domestic links are available in sizes 1/2" - 1.1/4". Working Load Limits, Dimensions, and Design Factors may vary by manufacturer.



IMPORTED AND DOMESTIC LINKS



PEAR SHAPED SLING LINKS

-D- Size in Inches	Working Load Limit in Pounds*	L x W1 x W2	Approximate Weight Each in Pounds
3/8	1,800	2.25 x 1.50 x .75	0.25
1/2	2,900	3.00 x 2.00 x 1.00	0.58
5/8	4,200	3.75 x 2.50 x 1.25	1.12
3/4	6,000	4.50 x 2.75 x 1.38	1.92
7/8	8,300	5.25 x 3.50 x 1.75	3.03
1	10,800	6.00 x 3.75 x 1.88	4.58
1.1/4	16,750	7.75 x 5.00 x 2.50	8.96
1.3/8	43,000	8.50 x 5.50 x 2.75	11.65
1.1/2	47,880	12.00 x 6.00 x 3.00	17.00
1.3/4	62,520	12.00 x 7.00 x 4.00	24.30
2	97,680	14.00 x 7.00 x 4.00	37.00
2.1/4	119,400	14.00 x 7.00 x 4.00	47.50
2.1/2	147,300	15.50 x 10.00 x 5.00	67.00
2.3/4	178,200	16.00 x 9.00 x 7.00	83.90

3/8" through 1.1/4" forged. Larger sizes welded. Heat treated, painted, or hot galvanized.

Sizes 3/8" thru 1.1/4" Carbon Steel, Ultimate Breaking Strength is 6 times Working Load Limit. Proof Load tested.

Sizes 1.3/8" thru 2.3/4" Alloy Steel. Ultimate Breaking Strength is 4 times Working Load Limit.

Proof Load Tested.

*Single Leg Sling (in-line) or maximum load capacity on multiple legs, where the included angle is 120° degrees or less.

Working Load Limits, Dimensions, and Design Factors may vary by manufacturer.

